

Biofuels

Learning outcomes

By the end of this section you should be able to:

- Summarise the reactants and products of photosynthesis.
- Contrast renewable and non-renewable energy sources.
- Recognise the advantages and disadvantages of biofuels.

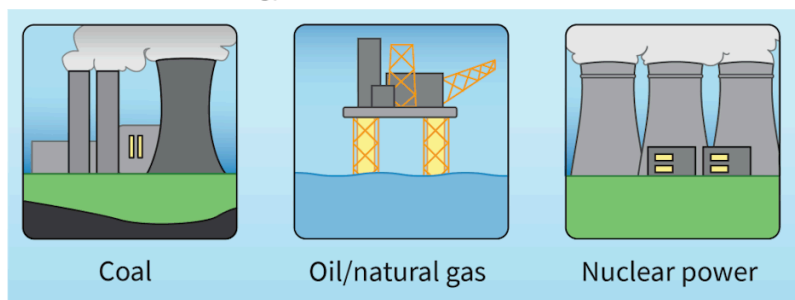
The Sun's energy is the driving force for nearly all energy cycles on Earth. For billions of years, life on Earth has evolved to best harness this practically limitless source of energy. Without the ability of plants, fungi, and other organisms to convert solar radiation to chemical energy, life on Earth would not have developed as we know it. The process by which living organisms convert solar energy to chemical energy is called photosynthesis. But how does this relate to chemistry? And how can we harness this limitless energy resource to meet our current and future energy requirements?

Renewable vs. non-renewable energy sources

All energy sources are classified as being renewable or non-renewable. A non-renewable energy source is consumed in a chemical or nuclear process and cannot be renewed at a rate equal to or faster than it is consumed. Energy resources like enriched uranium, which is used in fission nuclear reactors, and fossil fuels are non-renewable.

On the other hand, energy resources like solar energy, hydroelectricity, and wind power are classified as renewable. Other renewable resources like biofuels, and hydrogen fuel cells are more closely connected to chemical processes, and you will learn more about them in this section and section R1.3.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/>) respectively.

Non-renewable energy



Renewable energy

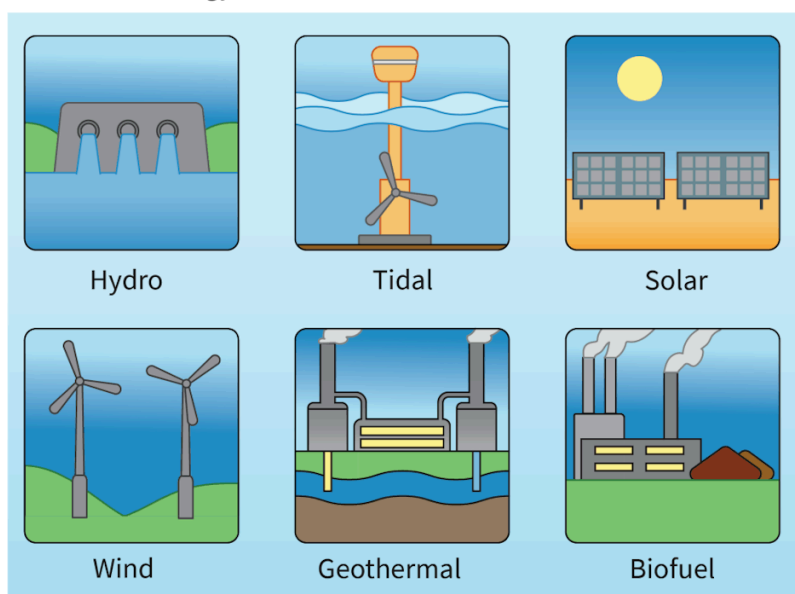


Figure 1. Renewable vs. non-renewable energy resources.

Renewable energy resources are often confused with green energy – energy sources that have a reduced environmental impact. In fact, both renewable and non-renewable energy resources have advantages and disadvantages much like the combustion of fossil fuels studied in [section R1.3.3](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fossil-fuels-id-45276/>). Use **Interactive 1** to sort the energy resources into renewable energy and non-renewable energy.

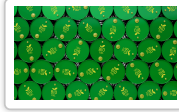
Renewable	Non-renewable











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Interactive 1. Renewable vs. Non-Renewable Energy Resources.

Photosynthesis and the production of biofuels

Biofuels are, as the name suggests, fuels that are derived from biological sources. These fuels, like those studied in [section R1.3.3](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fossil-fuels-id-45276/>), are organic molecules that can be combusted, and the energy released is harnessed for use in many different applications. What exactly is a biofuel, and how are these fuels formed?

To have a better understanding of biofuels we need to start at the beginning, which in this case is the Sun! The Sun emits electromagnetic radiation at various wavelengths, as seen in [section S1.3.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>), which corresponds to different levels of energy. Plants and other organisms harness this energy by absorbing certain wavelengths of light within their chlorophyll-containing pigments. The wavelengths of light that are absorbed by chlorophyll generate electrons to power photosynthesis.

The chlorophyll-containing pigments then produce glucose, which can be used by the plant for immediate energy or further transformed and stored. These photo-chemical reactions also require carbon dioxide gas and water vapour as the

primary reactants and produce oxygen gas as a by-product. This highly complex biological pathway, which consists of many different steps and mechanisms, is often reduced to a simple equation.

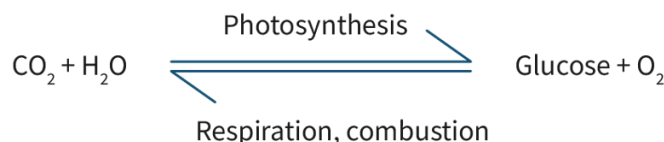


Figure 2. Photosynthesis and combustion.

 More information for figure 2

The diagram illustrates the processes of photosynthesis and combustion through two arrows. On the left side of the diagram, there are the chemical compounds "CO₂ + H₂O". An upward arrow is labeled "Photosynthesis," pointing from the left to the right, leading to "Glucose + O₂" on the right side. Below this, a downward arrow labeled "Respiration, combustion" points back from "Glucose + O₂" to "CO₂ + H₂O," showing the reverse process. This visual representation highlights the cyclic nature of these biological and chemical processes: carbon dioxide and water being converted into glucose and oxygen through photosynthesis, and then glucose and oxygen converting back into carbon dioxide and water through respiration or combustion.

[Generated by AI]

Some glucose molecules are used immediately by the plant for its energy needs, while others are converted to cellulose or polysaccharides like starch or into lipids to help with structure and energy storage. These molecules are not only used by plants for the storage of energy, but can be used by humans to generate energy as well. We do this by converting these molecules into combustible materials called biofuels. As the name implies, biofuels are fuels that are derived from biotic or living sources.

From plant material to fuels

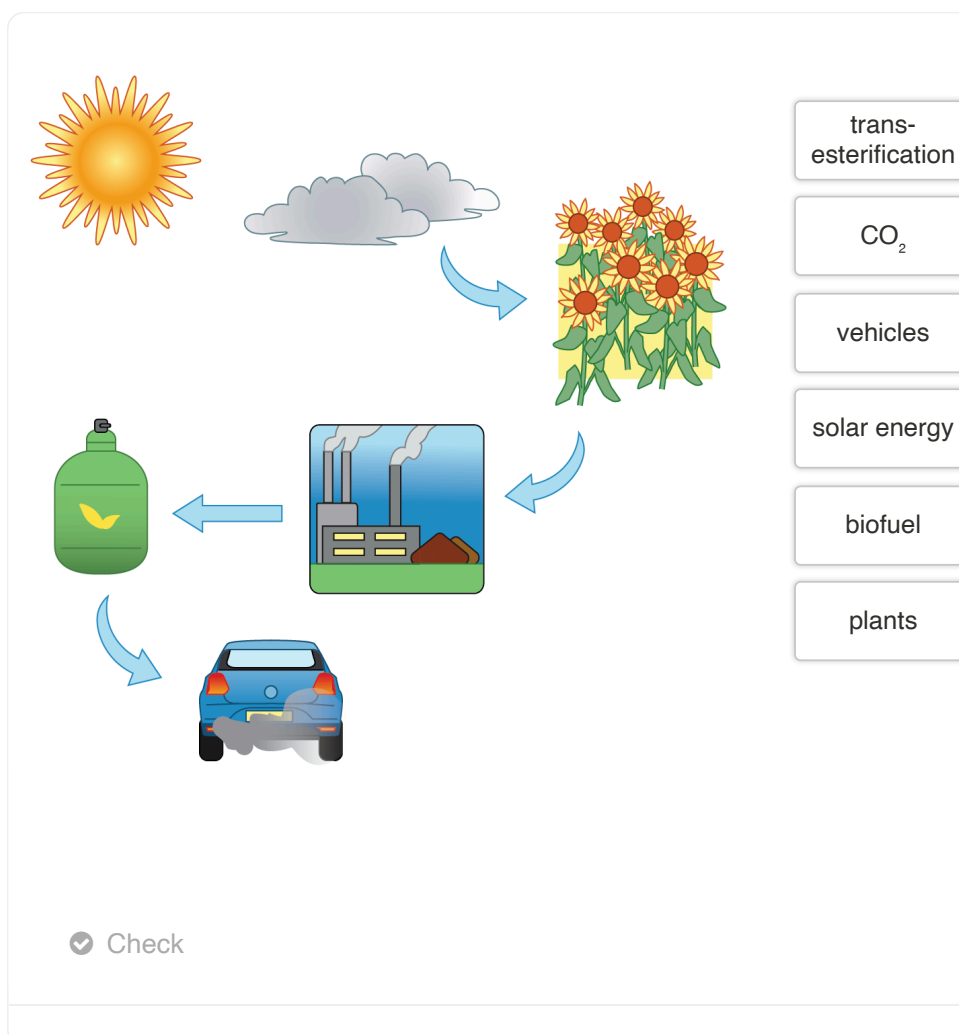
Biofuels are not a recent discovery and have been used as a form of energy by humans for thousands of years. If you think about it, humans have been burning plant-based oils and materials like wood since prehistoric times.

Modern day biofuels are derived from plant material that is transformed via physical and chemical processes to produce usable fuels. We then combust these biofuels in the presence of oxygen gas, producing carbon dioxide, water, and energy. The biofuels we use today, like ethanol and biodiesel, are more processed

and are derived from many different types of organisms; from sugar-rich crops like corn and sugar cane to cellulose, food, and animal waste, and more recent sources like algae and cyanobacteria.

No matter the primary source, the biofuels are produced via the breakdown or decomposition of larger organic molecules into combustible short-chain hydrocarbons like alkanes, alcohols, glycerol, and ethyl esters.

Use **Interactive 2** to complete the pathway representing the conversion of sunlight to biofuels.



Interactive 2. A pathway representing the conversion of sunlight to biofuels.

 More information for interactive 2

An interactive illustration displays the conversion of sunlight to biofuels using arrows to show each step in the cycle. Users need to drag and drop the correct terms into the appropriate drop zones that correspond to each stage of the process. The sequence in the illustration includes the following elements:

- The Sun
- Clouds
- Plants with flowers

- An industry/processing plant
- A green fuel can with two leaves
- A car with smoke coming out of the exhaust

The drag-and-drop options provided are:

Transesterification, CO₂, Vehicles, Solar energy, Biofuel, Plants

This interactivity helps users understand the transformation of solar energy into biofuels by engaging them in a visual and hands-on activity.

It introduces solar energy as the primary source and illustrates how plants absorb carbon dioxide and sunlight for growth through photosynthesis. Users learn about transesterification as the key chemical process that converts plant oils into usable biofuels. The final step connects biofuels to their real-world application in vehicles, highlighting the practical use of renewable energy.

Read below for the solution:

Sun- Solar energy

Clouds- CO₂

Plant with flowers- Plants

Industry- Transesterification

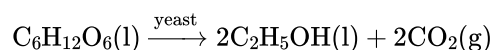
Green fuel can with two leaves - Biofuel

Car with smoke coming out of the exhaust- Vehicles

International Mindedness

Most countries, in their efforts to improve fuel security, have in place research and development programs for the production and distribution of biofuels.

In the latter half of the twentieth century, Brazil became one of the pioneers in the production of bioethanol. Bioethanol is a useful biofuel, which can be added to gasoline to produce flex fuels. Bioethanol is produced via the fermentation of plant material (feedstock) that is high in starch and sugars. This process takes place in anaerobic conditions in the presence of yeast, which act as a biological catalyst.



Many companies, around the globe, are also using food waste to produce biodiesel. This is achieved through the conversion of plant-based lipids (triglycerides) to glycerol and methyl ester through a reaction known as transesterification. The reaction needs to be carried out in the presence of a strong base and methanol or ethanol, and in the end old vegetable oil and grease can be converted into a usable fuel.

Advantages and disadvantages of biofuels

Biofuels have many advantages. The plants used to make biofuels must absorb carbon dioxide when performing photosynthesis, which offsets the carbon emissions of the biofuel and it undergoes combustion. As well, since some biofuels are short-chain hydrocarbons, the combustion reactions they undergo are

cleaner than traditional fossil fuels. As you may recall from earlier in this section, short-chain hydrocarbons undergo complete combustion more readily than long chain hydrocarbons.

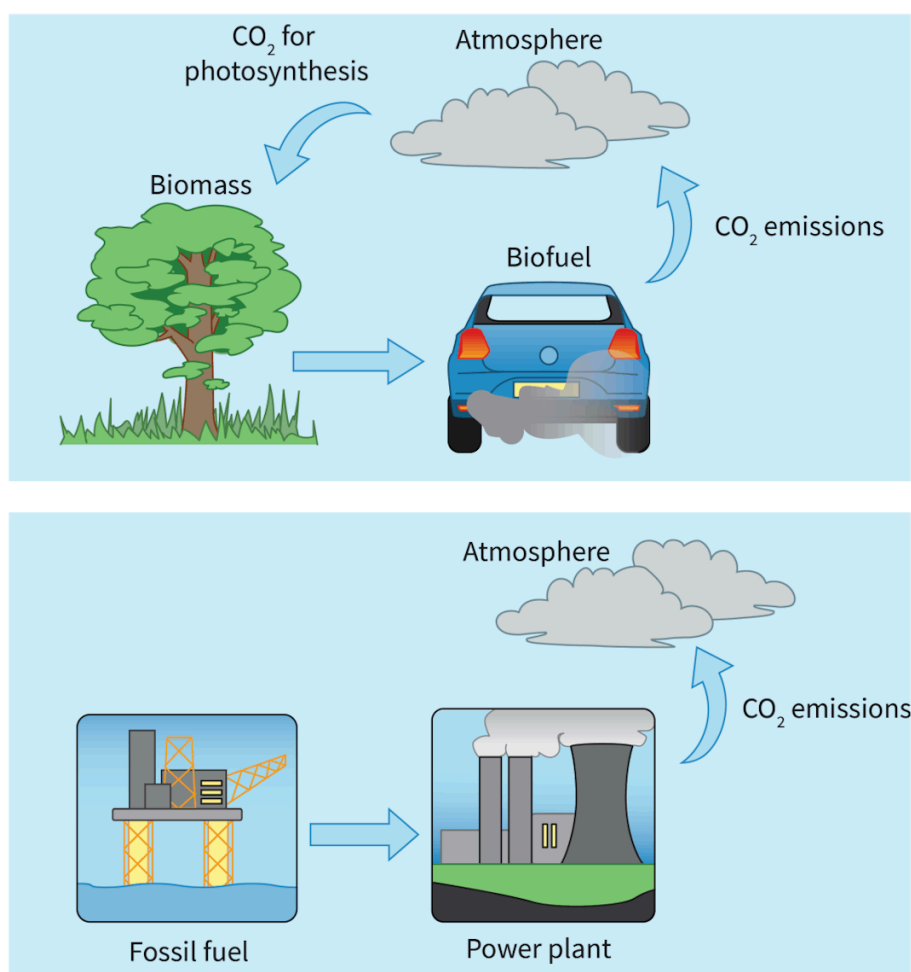


Figure 3. Partial offsetting with biofuels.

 More information for figure 3

The image is divided into two sections to illustrate the processes of biofuels and fossil fuels.

The top section shows the biofuel cycle. On the left, there is a graphic of a tree labeled "Biomass." An arrow points from the tree to a car labeled "Biofuel." This indicates the conversion of biomass into biofuel. The car emits "CO₂ emissions" depicted with an arrow pointing to a cloud labeled "Atmosphere." Another arrow circles back from the atmosphere to the tree with the text "CO₂ for photosynthesis," illustrating how plants absorb carbon dioxide, thus partially offsetting emissions.

The bottom section portrays the fossil fuel process. On the left, there is an image of an oil platform labeled "Fossil fuel." An arrow leads to a power plant on the right labeled "Power plant." Above the power plant, smoke is shown with an arrow pointing to another cloud labeled "Atmosphere," indicating "CO₂ emissions" from burning fossil fuels, which contribute to atmospheric pollution.

The diagram contrasts the carbon cycle in biofuels versus fossil fuels, emphasizing biofuels' less harmful environmental impact.

[Generated by AI]

In addition to this, the refining and synthesis process is less harmful to the environment, using fewer resources than fossil fuel refining. There is no need to mine these fuels or use fractional distillation in their production. Biofuels are relatively inexpensive to produce compared to other types of renewable energy resources, and the raw materials to make them are readily available. Lastly, by acting as a fossil fuel substitute, biofuels allow us to preserve our fossil fuel stores for future use in other applications.

On the other hand, biofuel production uses large quantities of plant material as its feedstock, which requires land that is appropriate for agriculture. Though the combustion of biofuels is somewhat less harmful than the combustion of traditional fossil fuels, the products produced in the combustion of biofuels still emit greenhouse gases into the atmosphere (**Table 1.**).

Table 1. Advantages and disadvantages of biofuels.

	Biofuels
Advantages	• Release less CO₂ compared to fossil fuels • Inexpensive • Can be used directly after synthesis — no mining or processing at a re as with fossil fuels
Disadvantages	• Uses land suitable for agriculture • Combustion still releases CO₂ into the atmosphere • Growth of crops requires the use of fossil fuels

Activity

- **IB learner profile attribute:**
 - Communicator
 - Thinker
- **Approaches to learning:**
 - Thinking skills — Evaluating and defending ethical positions
 - Communication skills — Clearly communicating complex ideas in response to open-ended questions, Using digital media for communicating information
- **Time required to complete activity:** 30–40 minutes

- **Activity type:** Individual activity

Instructions

In this section you explored the nature of renewable and non-renewable energy resources, and the origin of these sources of energy. In this activity you are encouraged to explore the production of biofuels in your region or country and to evaluate the ethical implications of investing in these types of fuel sources. Write a reflection on the process as well as the information you researched.

Many countries are turning to biofuels to meet their energy needs. Since the early 1970s Brazil has been investing heavily in the development of their biofuel industry.

1. Research what types of biofuels are produced in your region or country. What are the sources of these biofuels and how are they currently being distributed and used?
2. Evaluate what some of the benefits and drawbacks might be for a country investing in this type of energy resource. These benefits/drawbacks could be environmental, social, political, etc.